CO2 To Jet Fuel
Reverse engineering carbon dioxide and turning it into jet fuel; using organic combustion. **http://dgit.in/jan21-13**

Magnetocuring
Scientists invent new type of glue that gets activated by magnetic fields. **http://dgit.in/jan21-14**

Secrets of saliva

An apparently innocuous bodily secretion holds many secrets to our evolutionary past

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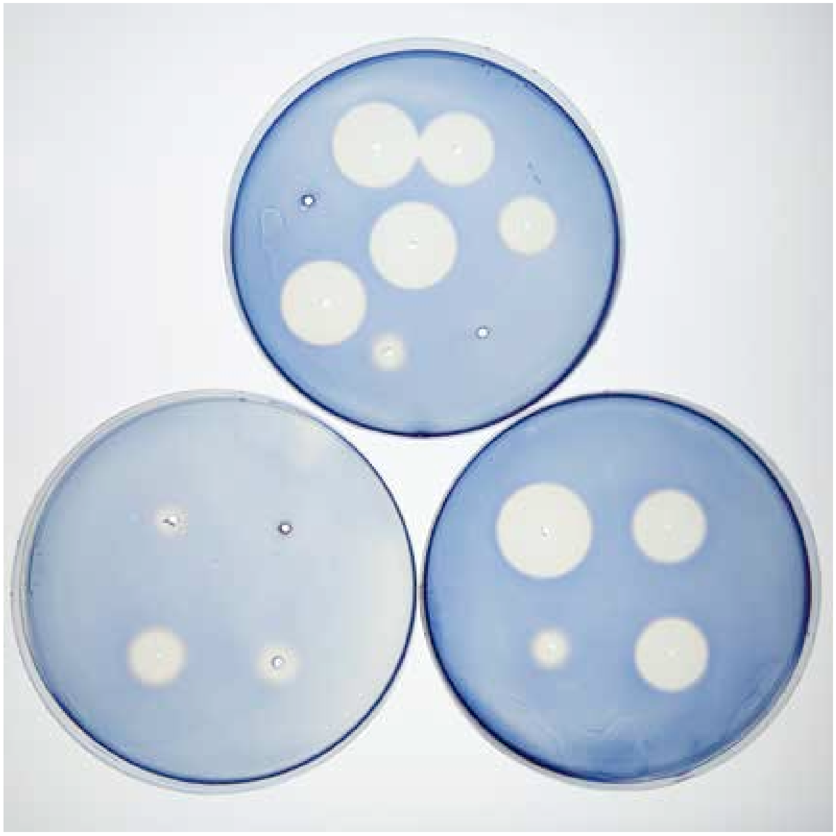
The saliva in humans and other animals is a complex mixture that is secreted by a number of glands and includes thousands of proteins. Omer Gokcumen and Stefan Ruhl from Buffalo University are the leading salivary researchers in the world, who have spent years researching the various proteins in saliva, the role of the various component proteins, and the exact mechanisms of its production.



Credit: Douglas Levere / University at Buffalo

The holes in the starchy gelatinous substrate have been introduced with saliva from various species of animals. The species with more amylase in the saliva show larger discs of starch breaking down.

THE ROLE OF AMYLASE

One of the studies compared the concentration and origin of a single enzyme in mammalian saliva, called amylase. The purpose of the study was to investigate how various animals evolved cellular machinery needed to produce the compound, and the relation of the diet of the various animals to the amount of amylase in the saliva of the species.

Across 46 mammals, it was found that the genetic machinery responsible for the production of amylase has evolved in many complex ways. Animals with starch rich diets have more copies of the gene that produces amylase, while animals with starch poor diets have fewer copies of the gene. While omnivores have more copies of the genes, animals that are primarily carnivores or insectivores have fewer. The researchers also found indications that the genetic machinery may have evolved independently, several times in the various organisms including in mice, dogs, pigs and humans. This indicates that the production of amylase is an example of convergent evolution, as well as possibly an example of the diet of animals can change their genetic makeup. One of the surprising findings of the study was that animals that were in close contact with humans or had access to human food, such as mice and dogs, had more copies of the amylase producing gene than their wild counterparts, rodents and wolves.

One of the interesting findings of the study was that the animals that produced the most amount of amylase among all the creatures tested were rhesus monkeys and baboons. Both of these species store food for long periods of time in their cheeks, during which time the amylase in the saliva helps break down the food. The exact advantage of producing amylase in animals that do not store food for long periods of time in special cheek pouches, are not exactly clear. For example, scientists still do not understand why humans need to produce the